

Name: _____ ID: _____ Section: _____

Height	Favourite Color	Gender	Weight	Ends in Nth Leaf Node
1.6	Blue	Female	Obese	1st
1.6	Red	Male	Not Obese	2nd
1.5	Red	Male	Obese	2nd
1.8	Red	Female	Not Obese	3rd
1.5	Green	Female	Obese	1st

Assume that the 1st leaf node has the value $(-0.6, 0.4)$; 2nd leaf node has the values $(0.4, -0.6)$ and the 3rd leaf node has the values (0.4) .

You are working on a gradient boosting classifier model. Now answer the following.

- Find the initial probability, and calculate the initial residual values. Consider the positive class is 1, and the negative class is 0. [5]
- What will be the probability of each instance after the first iteration of gradient boosting. [10]

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You are working on a gradient boosting classifier model. Now answer the following.

- c) Find the initial probability, and calculate the initial residual values. Consider the positive class is 1, and the negative class is 0. [5]
- d) What will be the probability of each instance after the first iteration of gradient boosting. [10]

1st-2nd : Assume that the 1st leaf node has the value $(-0.6, 0.4)$; 2nd leaf node has the values $(0.4, -0.6)$ and the 3rd leaf node has the values (0.4) .

1st-3rd: Assume that the 1st leaf node has the value $(-0.6, 0.4)$; 2nd leaf node has the values (-0.6) and the 3rd leaf node has the values $(0.4, 0.4)$.